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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: **B Tech/MBA Tech/B Tech (Integrated)** Year: **II/III/IV** Semester: **III/V/VII**

Stream/s ; IT, CE, CS, MXTC, EXTC, DS, AIDS, AIML, AI, CSE DS, CSE Cyber Sec

Subject: Data Structure and Algorithm Time: 3 hrs (10 .00 am to 1.00 pm)

Date: 4 / 12 / 2023 No. of Pages: 5

Marks: 100

Final Examination

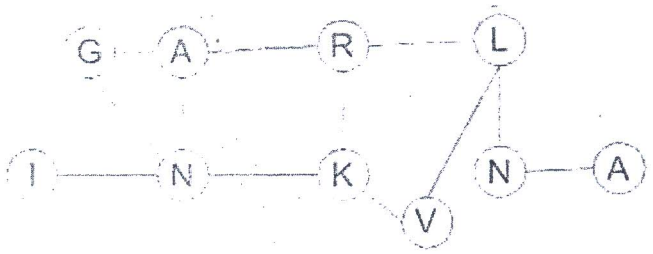
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

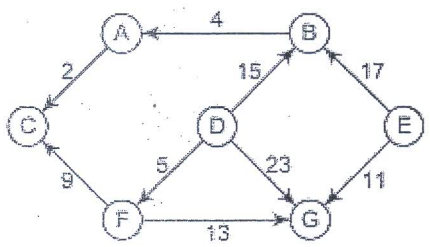
Q1		Answer briefly:	[20]
CO-1 ; SO-1,6; BL-4	a.	Compare and contrast the structural differences between linear data structures (e.g., arrays and linked lists) and hierarchical data structures (e.g., trees). Identify the most suitable data structure based on a given scenario: You are building a simple program to keep track of a to-do list. Each item on the list needs to be easily accessible for viewing, modification, and deletion. Which data structure would you choose and why?	5
CO-2 ; SO-1,6 ; BL-3	b.	Given a mathematical expression in infix notation, demonstrate the step-by-step process of converting it into polish notation. Use a stack to show how operands and operators are processed. (A+B)+C-(D-E)^F	5
CO-3 ; SO-1 ; BL-1,2	c.	A classroom contains 10 benches of which 7 are occupied as shown below. Draw the memory representation of linked list for storing the student's	5

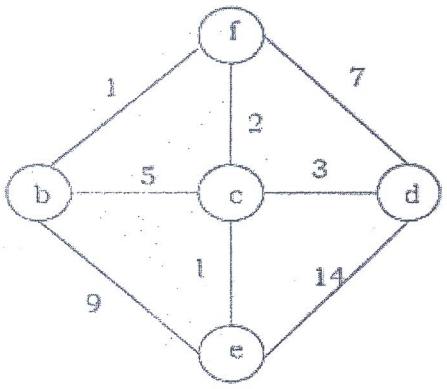
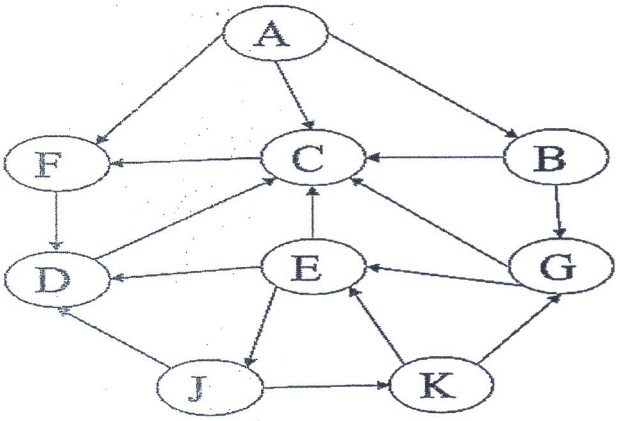
		names in alphabetical order. Draw the memory representation of linked list if student allocated bench number 3 and 7 leaves classroom in linked list created in I Draw the memory representation of linked list after allocating Roma bench number 8 in above linked list	

Bench Number	Student name
1	Kavita
2	
3	Diksha
4	Adil
5	Adnan
6	
7	Lara
8	
9	Sahil
10	Firoz

CO-3 ; SO-1 ; BL-1,2	d.	Define degree of a node in the context of a graph. Identify DFS sequence for below graph. Consider source node as G. Show each step.  	5
Q2 CO-1,2; SO-1,4; BL-2	(a)	Write a pseudo code function to insert an element in an integer array at a specified position by shifting elements to the right. Determine the time complexity in Big-O notation of the following function.	[10]

		int multiply(int A[], int N)
		{
		int s=1;
		for (int i=0; i< N; i++)
		s = s * A[i];
		return s;
		}	

	(b)	Write an algorithm to perform en-queue, de-queue and display operations on a circular queue using arrays. Check for full and empty queue conditions wherever required.	[10]
Q3 CO-3,; SO-1 ; BL-1,2	(a)	Apply selection sort on an array given and show each pass for the sorting. Index: 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 Values: 45 | 32 | 12 | 23 | 30 | 6 | 15 | 40 | 17	[10]
	(b)	Represent below graph as an adjacency matrix. Apply Dijkstra's algorithm for the following graph taking D as initial node. Show each step  	[10]
Q4 CO-3; SO-1,2; BL-3	(a)	Write an algorithm to perform insert operation in to binary search tree. Create a binary search tree by inserting the following elements in the given order. 60, 25, 72, 15, 30, 68, 101, 13, 18.	[10]
	(b)	Write an algorithm to perform in-order, post-order and pre-order traversal on binary tree.	[10]

		Construct Binary Tree from given traversals In-order sequence: 4, 8, 2, 5, 1, 6, 3, 7 Post-order sequence: 8, 4, 5, 2, 6, 7, 3, 1	
Q5 CO-3; SO-1; BL-2,3	(a)	Discuss two applications of minimum spanning tree. Represent below graph in a adjacency list form. Apply Kruskal' algorithm on below graph and identify the cost of connecting all the nodes. Show working at each step and final minimum spanning tree. 	[10]
	(b)	For the graph given below, apply BFS Algorithm to find all nodes reachable from node A. Show all the steps. Write an algorithm for the breadth first search traversal on graph. 	[10]
Q6 CO-2,4;	(a)	Explain different types of linked list. Write an algorithm for adding a new node at the end of a singly linked list and removing the first node from the	[10]

SO-1; BL- 2,3		singly linked list.	
	(b)	Write an algorithm for binary search. Perform binary search step-by-step on a given list for finding value 98. Show each step. 4, 21, 34, 40, 65, 76, 80, 92, 93, 97, 98	[10]
Q7 CO-2; SO-1; BL-3	(a)	Construct an expression tree for below expression. Identify prefix and postfix expression for the same. Show the working. A+(B/(D*F)-H)*T	[5]
CO-3; SO-1; BL-3	(b)	Construct the Huffman tree for the given symbols with their frequencies. Show each step. Identify the code for each letter. Ax= {a, b, c, d, e) Px= {30, 25, 20, 10,15}	[5]
CO-4; SO-1; BL-4	(c)	Consider a hash table with size=10. Using linear probing insert, the keys 27, 33, 64, 43, 37, 29, 30 and 102 into the table. Discuss the disadvantages of linear probing.	[10]